

ARYAN SAHOO

B.Tech Computer Science | Aspiring Software & Cloud Engineer

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SUMMARY

Third-year Computer Science student at VIT Vellore with hands-on project work in computer networks and operating systems. Building toward a career in cloud and backend engineering, with working knowledge of Java, Python, Spring Boot and core AWS services (EC2, S3, VPC, IAM). Enjoys understanding how systems work under the hood, from how networks route traffic to how an operating system schedules processes. Currently deepening AWS skills and building projects that connect coursework with real-world cloud infrastructure.

EDUCATION

Vellore Institute of Technology (VIT), Vellore

[2024 – 2028]

B.Tech in Computer Science and Engineering

CGPA: 7.89 / 10

CLOUD TRAINING

AWS Cloud Technical Essentials — Amazon Web Services (Coursera)

Sep 2026

- Covered AWS core services end to end: **compute** (EC2, Lambda, containers), **networking** (VPC, subnets, route tables, security groups), **storage** (S3, EBS) and **databases** (RDS, DynamoDB).
- Learned identity and security best practices with **IAM** users, roles and policies, plus the shared responsibility model.
- Studied designing highly available, scalable architectures using **Elastic Load Balancing**, **EC2 Auto Scaling** and monitoring with **CloudWatch**.
- Followed the architecture of a sample three-tier web application as it evolved across modules, from a single EC2 instance to a load-balanced, auto-scaled deployment.

PROJECTS

Predictive Digital-Twin-Assisted Traffic Engineering for SDN

Python, Mininet, Ryu/ONOS, NetworkX

Computer Networks course project | Team of 3 | Guide: Prof. Kopperundevi

- Designing a network digital twin that mirrors a software-defined network emulated in Mininet with a Ryu/ONOS controller.
- Integrating an **LSTM/TCN traffic predictor** with NetworkX-based path computation to reroute flows proactively before congestion occurs.
- Executing the build across a structured ~10-week phased plan.

SBACS — Security & Behaviour-Aware Adaptive CPU Scheduler

Operating Systems team project

- Own the **security behaviour analysis component**, which evaluates process behaviour and produces a per-process security-risk score.
- The risk score feeds the adaptive scheduler so that scheduling decisions account for security as well as performance.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C, C++, SQL, HTML/CSS

Frameworks & Tools: Spring Boot, Mininet, Ryu/ONOS, NetworkX

Cloud: AWS — EC2, S3, VPC, IAM, AWS databases, cloud security fundamentals

Areas of Interest: Cloud Computing, Backend Development, AI/ML, Computer Networks

CERTIFICATIONS

AWS Cloud Technical Essentials — Amazon Web Services (Coursera)

Sep 2026

Introduction to Networking — NVIDIA (Coursera)

Jun 2026